

Palo Alto Global Protect VPN Setup and Configuration



By Rui Yang



Background:

Cyberattacks, data loss, and identity theft is becoming both increasingly common and increasingly rare at the same time nowadays. As technology advances, more advanced attacking strategies are constantly being developed as well as defense mechanisms. This creates a continuous cycle of offense and defense in the cybersecurity field, where organizations must constantly evaluate and update their security posture to keep pace with possible attackers.

One of the most significant attacking angles in most organization is the remote network access. When employees or administrators need to connect to internal resources from outside the organization's physical network, that communication must travel across the public internet, where it is exposed to interception, manipulation, and unauthorized access. Without proper protection, sensitive data transmitted over these connections can be captured through techniques such as packet sniffing, where an attacker monitors network traffic to extract readable data, or man-in-the-middle attacks, where an attacker secretly positions themselves between two communicating parties to intercept or alter the data being exchanged.

Virtual Private Networks (VPNs) address this vulnerability by creating an encrypted tunnel between the user's device and the organization's internal network. All traffic passing through this tunnel is encrypted, meaning that even if an attacker manages to intercept the data in transit, it appears as a ciphered text without the corresponding decryption keys. This allows remote users to access internal resources as if they were physically present on the local network, without exposing that communication to the risks of the public internet.

However, VPNs differ from each other vastly. Consumer grade VPNs and basic remote access solutions often lack the advanced security features required in enterprise and institutional environments. Palo Alto's Next-Generation Firewalls (NGFWs) go beyond traditional firewall functionality by integrating deep packet inspection, application awareness, user identity tracking, and advanced threat prevention into a single platform. Palo Alto Networks' GlobalProtect is a VPN solution built directly into most of the Palo Alto NGFW, meaning that remote access traffic is not only encrypted but also subject to the same advanced security policies and threat inspection applied to all other traffic passing through the firewall. This integration ensures that

remote users are held to the same security standards as on-site users, reducing the possibility of remote network access attacks succeeding.

The Palo Alto PA-410 is an entry-level NGFW designed for small to medium sized deployments, offering the full feature set of Palo Alto's PAN-OS operating system in a compact and accessible shell. Configuring GlobalProtect on the PA-410 involves establishing the necessary certificates for encrypted communication, setting up a gateway that remote clients connect to, defining a portal that manages client configuration and distribution, and creating the security policies that govern what authenticated remote users are permitted to access once connected.

Purpose:

The purpose of this lab is to configure a GlobalProtect VPN solution on a Palo Alto PA-410 next-generation firewall, enabling secure encrypted remote access to an internal network. The configuration should allow authenticated users to establish a VPN tunnel from outside the network, with all traffic subject to the firewall's security policies upon connection.

Configurations:

Add a Root Certificate:

PA-410 DASHBOARD MONITOR POLICIES OBJECTS NETWORK **DEVICE**

Generate Certificate

Certificate Type: ☒ Local ☐ SCEP

Certificate Name:

Common Name:

IP or FQDN to appear on the certificate

Signed By:

☒ Certificate Authority

☐ Block Private Key Export

OCSP Responder:

Cryptographic Settings

Algorithm:

Number of Bits:

Digest:

Expiration (days):

Certificate Attributes

TYPE	VALUE
------	-------

Device Certificates | Default

NAME SUBJECT

DELETE REVOKE RENEW IMPORT **GENERATE** EXPORT CERTIFICATE IMPORT HA KEY

Add an intermediate certificate – make sure this is signed by the root cert:

The screenshot displays the Cisco Secure Manager interface. On the left is a navigation sidebar with categories like Setup, IoT Security, and Certificate Management. The main panel is titled 'Device Certificates' and contains a table with columns: NAME, SUBJECT, ISSUER, CA, KEY, and EXPIRES. A single entry, 'RootCert', is listed with SUBJECT 'CN = Root' and ISSUER 'CN = Root'. Overlaid on this is the 'Generate Certificate' dialog box. The dialog has two tabs: 'Local' (selected) and 'SCEP'. Fields include 'Certificate Name' (Intermediate-Cert), 'Common Name' (Intermediate), and 'Signed By' (RootCert). Checkboxes for 'Certificate Authority' and 'Block Private Key Export' are present. The 'Cryptographic Settings' section shows Algorithm: RSA, Number of Bits: 2048, Digest: sha256, and Expiration (days): 365. A 'Certificate Attributes' table is empty. At the bottom of the dialog are 'Generate' and 'Cancel' buttons. The bottom of the main window features a toolbar with icons for Delete, Revoke, Renew, Import, Generate, Export Certificate, Import HA Key, Export HA Key, and PDF/CSV.

NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES
RootCert	CN = Root	CN = Root			

Generate Certificate

Certificate Type: ☒ Local ☐ SCEP

Certificate Name: Intermediate-Cert

Common Name: Intermediate
IP or FQDN to appear on the certificate

Signed By: RootCert

☒ Certificate Authority
☐ Block Private Key Export

OCSP Responder:

Cryptographic Settings

Algorithm: RSA
Number of Bits: 2048
Digest: sha256
Expiration (days): 365

Certificate Attributes

TYPE	VALUE
------	-------

+ Add - Delete

Generate Cancel

Delete Revoke Renew Import Generate Export Certificate Import HA Key Export HA Key PDF/CSV

Add a Server certificate – make sure that this is signed by the intermediate cert:

The screenshot shows the PA-410 web interface with the 'Device Certificates' tab selected. A 'Generate Certificate' dialog box is open, allowing the user to create a new certificate. The dialog includes the following fields and options:

- Certificate Type:** Local (selected) or SCEP.
- Certificate Name:** Server-Cert
- Common Name:** 192.168.40.61 (IP or FQDN to appear on the certificate)
- Signed By:** Intermediate-Cert (dropdown menu)
- Options:** Certificate Authority (checkbox), Block Private Key Export (checkbox)
- OCSP Responder:** (dropdown menu)
- Cryptographic Settings:**
 - Algorithm: RSA
 - Number of Bits: 2048
 - Digest: sha256
 - Expiration (days): 365
- Certificate Attributes:** A table with columns TYPE and VALUE, and an 'Add' button.

The background shows the 'Device Certificates' table with the following data:

NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES
RootCert	CN = Root	CN = Root			
Intermedia...	CN = Intermediate	CN = Root			

Add a new SSL/TLS Service Profile:

The screenshot displays the PA-410 web interface with the 'DEVICE' tab selected. A modal window titled 'SSL/TLS Service Profile' is open, showing the configuration for a new service profile. The background interface includes a sidebar with various configuration options and a table with columns: NAME, LOCATION, CERTIFICATE, KEY EXCHANGE ALGORITHMS, PROTOCOL VERSIONS, and ENCR. The modal window contains the following settings:

- Name: SSL-TLS-Server
- Certificate: Server-Cert
- Protocol Settings:
 - Min Version: TLSv1.2
 - Max Version: TLSv1.3
- Key Exchange Algorithms:
 - ☒ RSA
 - ☒ DHE
 - ☒ ECDHE
- Encryption Algorithms:
 - ☒ AES-128-CBC
 - ☒ AES-256-CBC
 - ☒ AES-128-GCM
 - ☒ AES-256-GCM
 - ☒ AES-CHACHA20-POLY1305
- Authentication Algorithms:
 - ☒ SHA1
 - ☒ SHA256
 - ☒ SHA384

At the bottom of the modal, there are 'OK' and 'Cancel' buttons. The background interface also shows a bottom bar with 'Add', 'Delete', 'Clone', and 'PDF/CSV' options.

Make a Certificate Profile:

The screenshot displays the PA-410 web interface with the 'DEVICE' tab selected. The left sidebar shows the navigation menu, with 'Certificate Management' expanded and 'Certificate Profile' selected. The main content area shows a table with columns: NAME, LOCATION, USERNAME FIELD, USER DOMAIN, NAME, DEFAULT OCSP URL, OCSP VERIFY CERTIFICATE, USE CRL, USE OCSP, CRL RECEIVE TIMEOUT, OCSP RECEIVE TIMEOUT, and CERTIFICATE STATUS TIMEOUT. A 'Certificate Profile' configuration window is open, showing the following settings:

- Name: Client-Cert-Profile
- Username Field: None
- User Domain: (empty)
- CA Certificates: (empty table with columns: NAME, DEFAULT OCSP URL, OCSP VERIFY CERTIFICATE, TEMPLATE NAME/OID)
- Default OCSP URL (must start with http:// or https://): (empty)
- Use CRL: ☐ CRL Receive Timeout (sec): 5
- Use OCSP: ☐ OCSP Receive Timeout (sec): 5
- OCSP takes precedence over CRL: ☐ Certificate Status Timeout (sec): 5
- Block session if certificate status is unknown: ☐
- Block session if certificate status cannot be retrieved within timeout: ☐
- Block session if the certificate was not issued to the authenticating device: ☐
- Block sessions with expired certificates: ☐

At the bottom of the configuration window, there are 'OK' and 'Cancel' buttons. A smaller 'Certificate Profile' dialog box is also visible, showing the following settings:

- CA Certificate: RootCert
- Default OCSP URL: (empty)
- OCSP Verify Certificate: None
- Template Name/OID: (empty)

At the bottom of the dialog box, there are 'OK' and 'Cancel' buttons.

PA-410

DASHBOARD

MONITOR

POLICIES

OBJECTS

NETWORK

DEVICE

Setup

High Availability

Config Audit

Password Profiles

Administrators

Admin Roles

Authentication Profile

Authentication Sequence

User Identification

IoT Security

DHCP Server Log Ingestion

Data Redistribution

Device Quarantine

VM Information Sources

Troubleshooting

Certificate Management

Certificates

Certificate Profile

OCSP Responder

SSL/TLS Service Profile

SCEP

SSL Decryption Exclusion

SSH Service Profile

Response Pages

Log Settings

Server Profiles

Local User Database

Software

GlobalProtect Client

Dynamic Updates

Plugins

Licenses

Support

Master Key and Diagnostics

Policy Recommendation

IoT

SaaS

	NAME	LOCATION	USERNAME FIELD	USER DOMAIN	CA	DEFAULT OCSP URL	OCSP VERIFY CERTIFICATE	USE CRL	USE OCSP	CRL RECEIVE TIMEOUT	OCSP RECEIVE TIMEOUT	CERTIFICATE STATUS TIMEOUT
☒	Client-Cert-Profile				RootCert			☐	☐	5	5	5

Certificate Profile

Name

Client-Cert-Profile

Username Field

None

User Domain

CA Certificates

☐	NAME	DEFAULT OCSP URL	OCSP VERIFY CERTIFICATE	TEMPLATE NAME/OID
☐	RootCert			
☐	Intermediate-Cert			

+ Add

- Delete

↑ Move Up

↓ Move Down

Default OCSP URL (must start with http:// or https://)

☐ Use CRL

CRL Receive Timeout (sec)

5

☐ Use OCSP

OCSP Receive Timeout (sec)

5

OCSP takes precedence over CRL

Certificate Status Timeout (sec)

5

☐ Block session if certificate status is unknown

☐ Block session if certificate status cannot be retrieved within timeout

☐ Block session if the certificate was not issued to the authenticating device

☐ Block sessions with expired certificates

OK

Cancel

+ Add

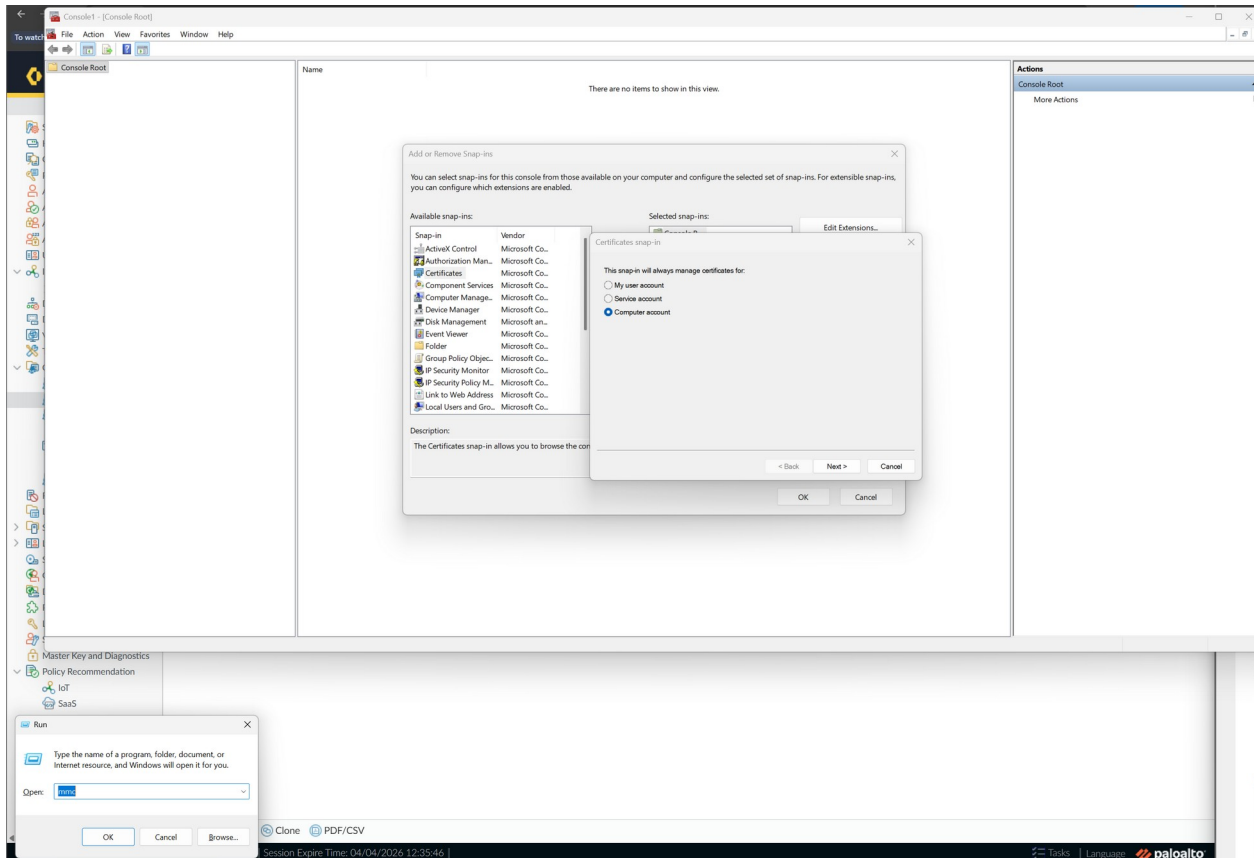
- Delete

Clone

PDF/CSV

admin | Logout | Last Login Time: 03/05/2026 11:35:36 | Session Expire Time: 04/04/2026 12:35:46 |

Optional: Add the Certificate on your end device through mmc. DO File -> Add or Remove Snap-in:



Add or Remove Snap-ins



You can select snap-ins for this console from those available on your computer and configure the selected set of snap-ins. For extensible snap-ins, you can configure which extensions are enabled.

Available snap-ins:

Snap-in	Vendor
ActiveX Control	Microsoft Co...
Authorization Man...	Microsoft Co...
Certificates	Microsoft Co...
Component Services	Microsoft Co...
Computer Manage...	Microsoft Co...
Device Manager	Microsoft Co...
Disk Management	Microsoft an...
Event Viewer	Microsoft Co...
Folder	Microsoft Co...
Group Policy Objec...	Microsoft Co...
IP Security Monitor	Microsoft Co...
IP Security Policy M...	Microsoft Co...
Link to Web Address	Microsoft Co...
Local Users and Gro...	Microsoft Co...

Add >

Selected snap-ins:

- Console Root
- Certificates (Local Computer)
- Certificates - Current User

Edit Extensions...

Remove

Move Up

Move Down

Advanced...

Description:

The Certificates snap-in allows you to browse the contents of the certificate stores for yourself, a service, or a computer.

OK

Cancel

Add an authentication profile:

The screenshot displays the PA-410 web interface. The top navigation bar includes 'DASHBOARD', 'MONITOR', 'POLICIES', 'OBJECTS', 'NETWORK', and 'DEVICE'. The left sidebar contains a tree view with categories like 'Setup', 'High Availability', 'Config Audit', 'Password Profiles', 'Administrators', 'Admin Roles', 'Authentication Profile' (highlighted with a red box), 'Authentication Sequence', 'User Identification', 'IoT Security', 'Data Redistribution', 'Device Quarantine', 'VM Information Sources', 'Troubleshooting', 'Certificate Management', 'Certificates', 'Certificate Profile', 'OCSP Responder', 'SSL/TLS Service Profile', 'SCEP', 'SSL Decryption Exclusion', 'SSH Service Profile', 'Response Pages', 'Log Settings', 'Server Profiles', 'SNMP Trap', 'Syslog', 'Email', 'HTTP', 'Netflow', 'RADIUS', 'SCP', 'TACACS+', 'LDAP', 'Kerberos', 'SAML Identity Provider', 'Multi Factor Authentication', 'Local User Database', 'Users', 'User Groups', 'Software', 'GlobalProtect Client', and 'Dynamic Updates'. The main content area shows a table with columns: NAME, LOCATION, LOCKOUT (FAILED ATTEMPTS (#), LOCKOUT TIME (MIN)), ALLOW LIST, AUTHENTICATION, and SERVER PROFILE. An 'Add' button (plus icon) is highlighted with a red box at the bottom left. A modal dialog titled 'Authentication Profile' is open, showing the 'Authentication' tab. The 'Name' field is 'Local_Auth'. The 'Type' dropdown is set to 'Local Database'. The 'User Domain' field is empty. The 'Username Modifier' dropdown is set to '%USERINPUT%'. The 'Single Sign On' section has 'Kerberos Realm' empty and 'Kerberos Keytab' with a button to 'Click "Import" to configure this field' and an 'X Import' link. 'OK' and 'Cancel' buttons are at the bottom right. The footer shows 'admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16'.

PA-410

DASHBOARD MONITOR POLICIES OBJECTS NETWORK DEVICE

Setup
High Availability
Config Audit
Password Profiles
Administrators
Admin Roles
Authentication Profile
Authentication Sequence
User Identification
IoT Security
DHCP Server Log Ingestion
Data Redistribution
Device Quarantine
VM Information Sources
Troubleshooting
Certificate Management
Certificates
Certificate Profile
OCSP Responder
SSL/TLS Service Profile
SCEP
SSL Decryption Exclusion
SSH Service Profile
Response Pages
Log Settings
Server Profiles
SNMP Trap
Syslog
Email
HTTP
Netflow
RADIUS
SCP
TACACS+
LDAP
Kerberos
SAML Identity Provider
Multi Factor Authentication
Local User Database
Users
User Groups
Software
GlobalProtect Client
Dynamic Updates

NAME LOCATION LOCKOUT (FAILED ATTEMPTS (#) LOCKOUT TIME (MIN) ALLOW LIST AUTHENTICATION SERVER PROFILE

Authentication Profile

Name Local_Auth

Authentication Factors Advanced

Type Local Database

User Domain

Username Modifier %USERINPUT%

Single Sign On

Kerberos Realm

Kerberos Keytab Click "Import" to configure this field X Import

OK Cancel

+ Add - Delete Clone PDF/CSV

admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16

Authentication Profile

NameLocal_Auth

Authentication

FactorsAdvanced

Allow List

☐ ALLOW LIST ^

☒ all

+ Add - Delete

Account Lockout

Failed Attempts[0 - 10]

Lockout Time (min)0

OKCancel

Create a VPN Tunnel:

The screenshot displays the PA-410 Network configuration interface. The top navigation bar includes 'DASHBOARD', 'MONITOR', 'POLICIES', 'OBJECTS', 'NETWORK' (highlighted), and 'DEVICE'. The left sidebar shows a tree view of configuration options, with 'Interfaces' highlighted. The main content area shows a table of interfaces with columns: INTERFACE, MANAGEMENT PROFILE, IP ADDRESS, VIRTUAL ROUTER, SECURITY ZONE, FEATURES, and COMMENT. A 'Tunnel' interface is listed with 'none' for IP ADDRESS, VIRTUAL ROUTER, and SECURITY ZONE. A 'Tunnel Interface' configuration dialog is open, showing fields for 'Interface Name' (tunnel), 'Comment', and 'Netflow Profile' (None). The 'Config' tab is selected, and the 'Assign Interface To' section shows 'Virtual Router' set to 'default' and 'Security Zone' set to 'Untrust_L3'. The 'Add' button in the bottom left is highlighted. The bottom status bar shows 'admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16 |'.

PA-410

DASHBOARD MONITOR POLICIES OBJECTS NETWORK DEVICE

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Portals

Gateways

MDM

Clientless Apps

Clientless App Groups

DHCP Profile

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

Ethernet VLAN Loopback Tunnel SD-WAN

INTERFACE	MANAGEMENT PROFILE	IP ADDRESS	VIRTUAL ROUTER	SECURITY ZONE	FEATURES	COMMENT
tunnel		none	none	none		

Tunnel Interface

Interface Name: tunnel . 10

Comment:

Netflow Profile: None

Config IPv4 IPv6 Advanced

Assign Interface To

Virtual Router: default

Security Zone: Untrust_L3

OK Cancel

+ Add - Delete PDF/CSV

admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16 |

Create a GlobalProtect Portal:

The screenshot displays the Palo Alto Networks PA-410 management interface. The top navigation bar includes 'DASHBOARD', 'MONITOR', 'POLICIES', 'OBJECTS', 'NETWORK' (highlighted with a red box), and 'DEVICE'. On the left sidebar, the 'GlobalProtect' section is expanded, and the 'Portals' option is highlighted with a red box. The main content area shows a table with columns: NAME, LOCATION, INTERFACE, IP, SSL/TLS SERVICE PROFILE, and AUTHENTICATION PROFILE. A 'GlobalProtect Portal Configuration' dialog box is open, featuring a left-hand menu with 'General' (highlighted with a red box), 'Authentication', 'Portal Data Collection', 'Agent', 'Clientless VPN', and 'Satellite'. The 'General' tab contains the following settings:

- Name:** GP-Portal (highlighted with a red box)
- Network Settings:**
 - Interface:** ethernet1/1 (highlighted with a red box)
 - IP Address Type:** IPv4 Only
 - IPv4 Address:** None
- Appearance:**
 - Portal Login Page:** factory-default
 - Portal Landing Page:** factory-default
 - App Help Page:** None
- Log Settings:**
 - ☐ Log Successful SSL Handshake
 - ☒ Log Unsuccessful SSL Handshake
 - Log Forwarding:** None

At the bottom right of the dialog are 'OK' and 'Cancel' buttons. The bottom status bar shows 'admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16 |'.

PA-410

DASHBOARDMONITORPOLICIESOBJECTSNETWORKDEVICE

Interfaces

Zones

VLANs

Virtual Wires

Virtual Routers

IPSec Tunnels

GRE Tunnels

DHCP

DNS Proxy

GlobalProtect

Gateways

MDM

Clientless Apps

Clientless App Groups

DHCP Profile

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

BFD Profile

SD-WAN Interface Profile

NAMELOCATIONINTERFACEIPSSL/TLS SERVICE PROFILEAUTHENTICATION PROFILE

GlobalProtect Portal Configuration

General

Authentication

Portal Data Collection

Agent

Server Authentication

SSL/TLS Service ProfileSSL-TLS-Server

Client Authentication

Client Authentication

NameLocal Authentication

OSAny

Authentication ProfileLocal_Auth

Automatically retrieve passcode from SoftToken application

Use default browser

GlobalProtect App Login Screen

Username LabelUsername

Password LabelPassword

Authentication MessageEnter login credentials

Authentication message can be up to 256 characters.

Allow Authentication with User Credentials OR Client Certificate

Yes (User Credentials OR Client Certificate Required)

To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.

OK

Cancel

AddDeleteGenerate TicketPDF/CSV

admin | Logout | Last Login Time: 03/05/2026 11:35:46 | Session Expire Time: 04/05/2026 12:22:16 |

Edit the Agent configuration:

GlobalProtect Portal Configuration

General

Authentication

Portal Data Collection

Agent

Clientless VPN

Satellite

Agent

CONFIGS

USER/USER GROUP

OS

EXTERNAL GATEWAYS

CLIENT CERTIFICATE

+ Add

- Delete

Clone

↑ Move Up

↓ Move Down

TRUSTED ROOT CA

RootCert

INSTALL IN LOCAL ROOT CERTIFICATE STORE

+ Add

- Delete

Agent User Override Key

Confirm Agent User Override Key

OK

Cancel

Configs

Authentication

Config Selection Criteria

Internal

External

App

HIP Data Collection

Name

GP-client-config-1

Client Certificate

None

The selected client certificate including its private key will be installed on client machines.

Save User Credentials

Yes

Authentication Override

Generate cookie for authentication override

Accept cookie for authentication override

Cookie Lifetime

Hours

24

Certificate to Encrypt/Decrypt Cookie

RootCert

Components that Require Dynamic Passwords (Two-Factor Authentication)

Portal

Internal gateways-all

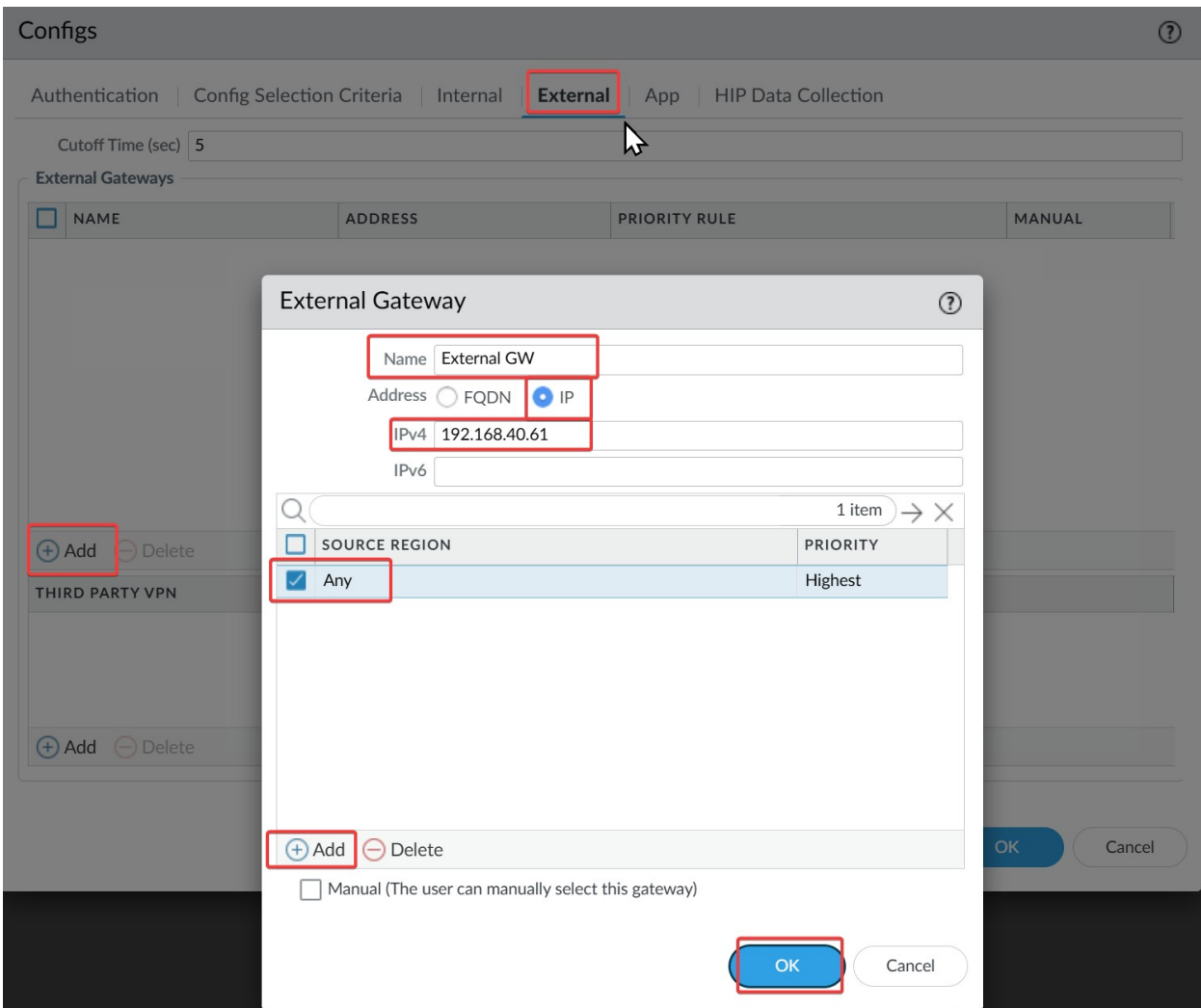
External gateways-manual only

External gateways-auto discovery

Select the options that will use dynamic passwords like one-time password (OTP) to authenticate users as opposed to using saved credentials. As a result, the user will always be prompted to enter new credentials for each selected option.

OK

Cancel



Configs



Authentication

Config Selection Criteria

Internal

External

App

HIP Data Collection

App Configurations

Connect Method	Pre-logout (Always On)
GlobalProtect App Config Refresh Interval (hours)	24 [1 - 168]
Allow user to disconnect GlobalProtect App (Always-on mode)	Allow with Ticket
Display the following reasons to disconnect GlobalProtect (Always-on mode)	
Allow User to Uninstall GlobalProtect App (Windows Only)	Allow
Allow User to Upgrade GlobalProtect App	Allow with Prompt
Allow user to Sign Out from GlobalProtect App	Yes
Allow user to extend GlobalProtect User Session	No

Welcome Page **None**

Disconnect GlobalProtect App (Always-on mode)

Passcode

Confirm Passcode

Max Times User Can Disconnect

Disconnect Timeout (min)

Uninstall GlobalProtect App

Uninstall Password

Confirm Uninstall Password

Mobile Security Manager Settings

Mobile Security Manager

Enrollment Port

OK

Cancel

Create and Configure a GlobalProtect Gateway:

The screenshot displays the Palo Alto Networks PA-410 configuration interface. The left sidebar shows the navigation menu with 'Gateways' highlighted. The main panel shows the 'GlobalProtect Gateway Configuration' dialog box. The 'General' tab is active, showing the following settings:

- Name: GlobalGateway
- Interface: ethernet1/1
- IP Address Type: IPv4 Only
- IPv4 Address: None
- Log Settings:
 - ☐ Log Successful SSL Handshake
 - ☒ Log Unsuccessful SSL Handshake
- Log Forwarding: None

The 'Add' button at the bottom left of the main panel is highlighted with a red box.

GlobalProtect Gateway Configuration

General

Authentication

Agent

Satellite

Server Authentication

SSL/TLS Service Profile SSL-TLS-Server

Client Authentication

NAME OS

+ Add Delete Clone

Certificate Profile None

Block

Client Authentication

Name test

OS Any

Authentication Profile Local_Auth

Automatically retrieve passcode from SoftToken application

GlobalProtect App Login Screen

Username Label Username

Password Label Password

Authentication Message Enter login credentials

Authentication message can be up to 256 characters.

Allow Authentication with User Credentials OR Client Certificate No (User Credentials AND Client Certificate Required)

To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.

OK Cancel

Configure Agent Settings:

GlobalProtect Gateway Configuration

General

Authentication

Agent

Satellite

Tunnel Settings Client Settings Client IP Pool Network Services Connection Settings Video Traffic HIP Notific

Tunnel Mode

Tunnel Interface tunnel.10

Max User [1 - 250]

Enable IPSec

GlobalProtect IPSec Crypto default

Enable X-Auth Support

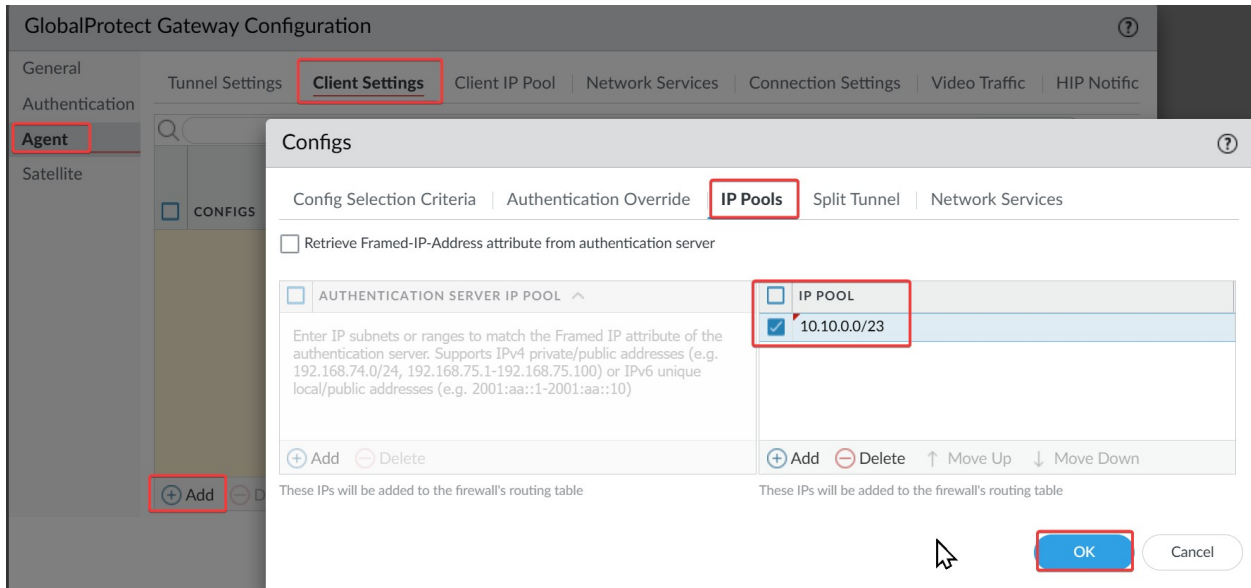
Group Name

Group Password

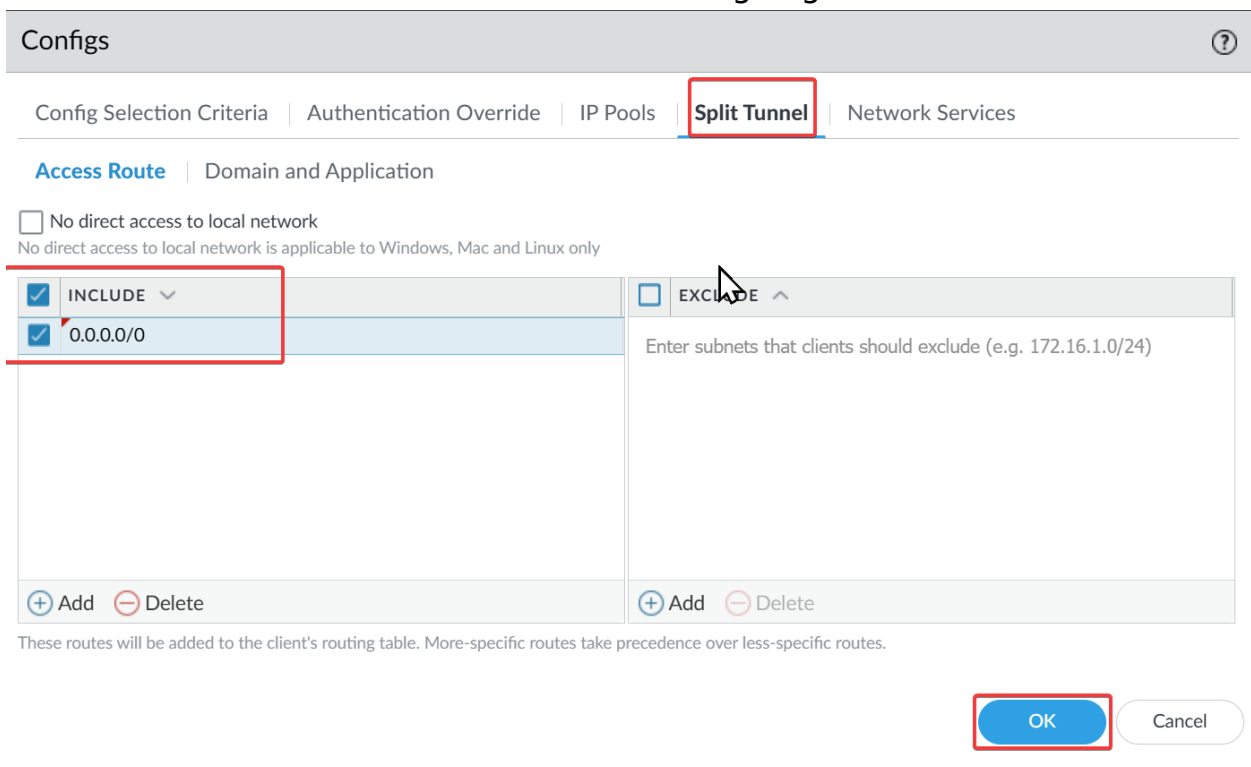
Confirm Group Password

Skip Auth on IKE Rekey

OK Cancel



Specify the route that traffic would route with, specifically use 0.0.0.0/0 to route all traffic to the Palo Alto Firewall before going outwards.



Commit Changes.

Problems:

The certificates had to be set to a specific ip address, and as the firewall's WAN facing address was a dynamic address, it would sometimes change, forcing us to delete the entire certification chain and remake the entire chain.

Remember to commit changes occasionally as you make edits, various issues were encountered because configurations were not saved and sometimes the firewall would restart for reasons unknown and progress would be lost and time would be spent troubleshooting why.

Conclusion:

In this lab, we successfully configured GlobalProtect VPN on a Palo Alto PA-410 Firewall. The configuration process began with establishing a certificate chain of trust, starting with a root certificate, followed by an intermediate certificate signed by the root, and finally a server certificate signed by the intermediate. This certificate chain ensures that client devices can verify the authenticity of the firewall before establishing a connection, reducing/preventing impersonation attacks. An SSL/TLS Service Profile and Certificate Profile were then created to define how the firewall presents and validates these certificates during the handshake process.

After that, an authentication profile was configured to control how users verify their identity before being granted VPN access. A dedicated VPN tunnel interface was then created to act as a path for the encrypted remote traffic. Then we configured a GlobalProtect Portal and configured its' agent settings, establishing the entry point that remote devices would connect to to receive their configuration and be directed to the appropriate gateway. Then the GlobalProtect Gateway was set up and configured with it's own set of agent settings and a tunnel route of 0.0.0.0/0, which forces all client traffic through the Palo Alto firewall before reaching the internet or internal resources, ensuring that no traffic bypasses the firewall's security inspection regardless of its destination.

Finally, the setup was confirmed by a simulation of a real possible event. We connected to the VPN from a device outside the local network and confirmed that the tunnel was established successfully, then used Remote Desktop from the external device to connect to a machine inside the local network, and the firewall's web GUI and CLI were both accessed successfully through

this connection, confirming that the VPN was correctly routing traffic and that internal resources were fully reachable through the encrypted tunnel as intended.

Optionally, the certificate can also be installed on the end device through the Microsoft Management Console to establish local trust to prevent certificate warnings during connection.